

Dataset Description

IIT-H Campus Life Prediction

The objective of this dataset is to model and predict **mess duration (in minutes)** — the amount of time a student from IITH spends in the mess during a particular meal — using contextual, environmental, and lifestyle-related features.

1 Dataset Overview

The dataset consists of 500 base samples collected over a simulated 15-day period, each sample corresponds to one student visiting the mess for a particular meal. 5000 extended samples created using Gaussian noise injection

2 Feature Description

2.1 Student and Temporal Features

Student IDs have to be in IITH format combining branch, admission year ,and roll number.

- Branch : ai, cs, me, bt, ee, ch, ce, ma
- Admission year : 22,23,24,25
- Roll numbers:11001-11050

Feature	Description
student_id	random unique IITH style id
day_of_week	Integer in [0,6], Monday = 0
meal_time	0 = breakfast, 1 = lunch, 2 = dinner
meal_type	0 = vegetarian, 1 = non-vegetarian

2.2 Binary Features

Feature	Description
is_weekend	1 if Saturday/Sunday, else 0
class_day	0 if classes present, 1 otherwise
assignment_deadline	1 if deadline present with probabiltly 0.2, else 0

2.3 Weather Features

Weather variables are generated using the following distributions:

- Temperature: $\mathcal{N}(30, 5)$
- Humidity: Uniform[40, 90]
- Windspeed: Uniform[0, 20]
- Rain: 70% probability of 0, otherwise Uniform[1, 20]
- Air Quality Index (AQI): Uniform[50, 200]

2.4 Personal Routine Features

- Rising time: Random integer between 300 and 600 minutes from midnight
- Sleeping time: Random integer between 1320 and 1440 minutes from midnight

3 Target Variable Construction

The target variable is:

$$\text{mess_duration} \quad (\text{minutes})$$

The target is constructed using a structured behavioral model.

Step 1: Base Duration by Meal

- Breakfast: 15–30 minutes
- Lunch: 30–50 minutes
- Dinner: 35–60 minutes

Step 2: Contextual Adjustments

- Weekend: +5 to +15 minutes
- Assignment deadline: -5 to -10 minutes
- Heavy rain (>10 mm): +5 minutes
- Poor air quality (AQI >150): -3 minutes

The final duration is clipped to a minimum of 10 minutes.

Thus, mess duration is computed as:

$$\text{Duration} = \text{BaseMealTime} + \text{ContextAdjustments}$$

This formulation introduces nonlinear dependencies between features and the target.

4 Gaussian Noise Augmentation

To simulate real-world measurement variability:

- The 500-sample dataset is replicated 10 times.
- Gaussian noise $\mathcal{N}(0, 2)$ is added to:
 - temperature
 - humidity
 - windspeed
 - airquality
 - rising_time
 - sleeping_time

And gaussian noise $\mathcal{N}(0, 1)$ is added to rain

This produces a final dataset of 5000 samples while preserving structural relationships.